
Quantitative Seismology Assignment 2

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Assignment: Number 2

Problem 1. Properties of Green's Function Using the elastodynamic equation (2.36) satisfied by Green's function and the reciprocity theorem (2.35).

(1) Prove $G(\mathbf{x}, t; \boldsymbol{\xi}, \tau) = G(\mathbf{x}, t - \tau; \boldsymbol{\xi}, 0) = G(\mathbf{x}, -\tau; \boldsymbol{\xi}, -t)$

The elastodynamic equation (2.36) for Green's function:

$$\rho \frac{\partial^2}{\partial t^2} G_{in}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) = \delta_{in} \delta(\mathbf{x} - \boldsymbol{\xi}) \delta(t - \tau) + \frac{\partial}{\partial x_j} \left(c_{ijkl} \frac{\partial}{\partial x_l} G_{kn}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) \right) \quad (1)$$

Define the operator $\mathcal{L}_{ik}$ acting on G_{kn} :

$$\mathcal{L}_{ik} \equiv \delta_{ki} \rho \frac{\partial^2}{\partial t^2} - \frac{\partial}{\partial x_j} \left(c_{ijkl} \frac{\partial}{\partial x_l} \right) \quad (2)$$

Then (1) becomes:

$$\mathcal{L}_{ik} G_{kn}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) = \delta_{in} \delta(\mathbf{x} - \boldsymbol{\xi}) \delta(t - \tau) \quad (3)$$

Shift t and τ by Δt in (1):

$$\mathcal{L}_{ik} G_{kn}(\mathbf{x}, t + \Delta t; \boldsymbol{\xi}, \tau + \Delta t) = \delta_{in} \delta(\mathbf{x} - \boldsymbol{\xi}) \delta(t - \tau) \quad (4)$$

Subtracting (3) from (4):

$$\mathcal{L}_{ik} (G_{kn}(\mathbf{x}, t + \Delta t; \boldsymbol{\xi}, \tau + \Delta t) - G_{kn}(\mathbf{x}, t; \boldsymbol{\xi}, \tau)) = 0 \quad (5)$$

Since $\mathcal{L}(u) = 0$ and let the Green's function of $\mathcal{L}$ be $g(\mathbf{x}, t; \boldsymbol{\xi}, \tau)$, then by spatiotemporal convolution:

$$u_i(\mathbf{x}, t) = \int_{-\infty}^{\infty} d\tau \iiint_V g_{ij}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) f_j(\boldsymbol{\xi}, \tau) dV(\boldsymbol{\xi}) = (g * f)_i \quad (6)$$

With $f_j = 0$, we have $u_i = (g * 0)_i = 0$. By the uniqueness theorem (see Aki & Richards, Chapter 2.3.1, strain-energy function), $u = 0$ is the only solution of $\mathcal{L}(u) = 0$. Thus from (5):

$$G_{kn}(\mathbf{x}, t + \Delta t; \boldsymbol{\xi}, \tau + \Delta t) - G_{kn}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) = 0 \quad (7)$$

Take $\Delta t = -\tau$ in (7):

$$G_{kn}(\mathbf{x}, t - \tau; \boldsymbol{\xi}, 0) = G_{kn}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) \quad (8)$$

Take $\Delta t = -t - \tau$ in (7):

$$G_{kn}(\mathbf{x}, -\tau; \boldsymbol{\xi}, -t) = G_{kn}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) \quad \text{Done (1)} \quad (9)$$

And by a similar way, we can also prove that shifting $\Delta \mathbf{x}$ in space won't change the Green's function, i.e.:

$$G_{kn}(\mathbf{x} + \Delta \mathbf{x}, t; \boldsymbol{\xi} + \Delta \mathbf{x}, \tau) = G_{kn}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) \quad (10)$$

Therefore, the Green's function depends only on the relative position:

$$G_{kn}(\mathbf{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = G_{kn}(\mathbf{x} - \boldsymbol{\xi}, t - \boldsymbol{\tau}; 0, 0) \quad (11)$$

Then equation (6) becomes the standard convolution form:

$$u_i(\mathbf{x}, t) = \int_{-\infty}^{\infty} d\boldsymbol{\tau} \iiint_V g_{ij}(\mathbf{x} - \boldsymbol{\xi}, t - \boldsymbol{\tau}) f_j(\boldsymbol{\xi}, \boldsymbol{\tau}) dV(\boldsymbol{\xi}) = (g * f)_i \quad (12)$$

(2) Prove $G_{nm}(\mathbf{x}, t; \boldsymbol{\xi}, 0) = G_{mn}(\boldsymbol{\xi}, t; \mathbf{x}, 0)$

The reciprocity theorem (2.35):

$$\begin{aligned} & \int_{-\infty}^{\infty} dt \iiint_V (u(\mathbf{x}, t)g(\mathbf{x}, \boldsymbol{\tau} - t) - v(\mathbf{x}, \boldsymbol{\tau} - t)f(\mathbf{x}, t)) dV \\ &= \int_{-\infty}^{\infty} dt \iint_S (v(\mathbf{x}, \boldsymbol{\tau} - t) \cdot T(u(\mathbf{x}, t)) - u(\mathbf{x}, t) \cdot T(v(\mathbf{x}, \boldsymbol{\tau} - t))) dS \end{aligned} \quad (13)$$

Use the homogeneous boundary condition: v , u or T equal to 0 at S .

Then (13) becomes:

$$\int_{-\infty}^{\infty} dt \iiint_V u(\mathbf{x}, t)g(\mathbf{x}, \boldsymbol{\tau} - t) dV = \int_{-\infty}^{\infty} dt \iiint_V v(\mathbf{x}, \boldsymbol{\tau} - t)f(\mathbf{x}, t) dV \quad (14)$$

Take $f(\mathbf{x}, t) = \delta_{in}\delta(\mathbf{x} - \boldsymbol{\xi}_1)\delta(t - t_1)$, $g(\mathbf{x}, t) = \delta_{im}\delta(\mathbf{x} - \boldsymbol{\xi}_2)\delta(t - t_2)$.

Then $u_i(\mathbf{x}, t) = G_{in}(\mathbf{x}, t; \boldsymbol{\xi}_1, t_1)$, $v_i(\mathbf{x}, t) = G_{im}(\mathbf{x}, t; \boldsymbol{\xi}_2, t_2)$.

Substituting into (14):

$$\begin{aligned} & \int_{-\infty}^{\infty} dt \iiint_V G_{in}(\mathbf{x}, t; \boldsymbol{\xi}_1, t_1)\delta_{im}\delta(\mathbf{x} - \boldsymbol{\xi}_2)\delta(\boldsymbol{\tau} - t - t_2) dV \\ &= \int_{-\infty}^{\infty} dt \iiint_V G_{im}(\mathbf{x}, \boldsymbol{\tau} - t; \boldsymbol{\xi}_2, t_2)\delta_{in}\delta(\mathbf{x} - \boldsymbol{\xi}_1)\delta(t - t_1) dV \end{aligned} \quad (15)$$

Evaluating the integrals in (15):

$$G_{mn}(\boldsymbol{\xi}_2, \boldsymbol{\tau} - t_2; \boldsymbol{\xi}_1, t_1) = G_{nm}(\boldsymbol{\xi}_1, \boldsymbol{\tau} - t_1; \boldsymbol{\xi}_2, t_2) \quad (16)$$

By (8), L.H.S. of (16) shift $\boldsymbol{\tau} - t_2, t_1$ by $-t_1$:

$$G_{mn}(\boldsymbol{\xi}_2, \boldsymbol{\tau} - t_2 - t_1; \boldsymbol{\xi}_1, 0) \quad (17)$$

R.H.S. of (16) shift $\boldsymbol{\tau} - t_1, t_2$ by $-t_2$:

$$G_{nm}(\boldsymbol{\xi}_1, \boldsymbol{\tau} - t_1 - t_2; \boldsymbol{\xi}_2, 0) \quad (18)$$

Let $\boldsymbol{\xi}_2 = \boldsymbol{\xi}$, $\boldsymbol{\xi}_1 = \mathbf{x}$, $\boldsymbol{\tau} - t_1 - t_2 = t$ in (17) and (18):

$$G_{mn}(\boldsymbol{\xi}, t; \mathbf{x}, 0) = G_{nm}(\mathbf{x}, t; \boldsymbol{\xi}, 0) \quad (19)$$

(3) Prove $G_{nm}(\mathbf{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = G_{mn}(\boldsymbol{\xi}, -\boldsymbol{\tau}; \mathbf{x}, -t)$

Replace t by $t - \boldsymbol{\tau}$ in (19):

$$G_{mn}(\boldsymbol{\xi}, t - \boldsymbol{\tau}; \mathbf{x}, 0) = G_{nm}(\mathbf{x}, t - \boldsymbol{\tau}; \boldsymbol{\xi}, 0) \quad (20)$$

L.H.S. of (20) shift $t - \boldsymbol{\tau}, 0$ by $-t$:

$$G_{mn}(\boldsymbol{\xi}, -\boldsymbol{\tau}; \mathbf{x}, -t) \quad (21)$$

R.H.S. of (20) shift $t - \boldsymbol{\tau}, 0$ by $\boldsymbol{\tau}$:

$$G_{nm}(\mathbf{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) \quad (22)$$

Then from (21) and (22):

$$G_{nm}(\mathbf{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = G_{mn}(\boldsymbol{\xi}, -\boldsymbol{\tau}; \mathbf{x}, -t) \quad (23)$$

Problem 2. Conservation of Total Momentum $f^{[u]}$ is body-force equivalent for displacement discontinuity across fault plane.

$[u_i(\boldsymbol{\xi}, \boldsymbol{\tau})] \triangleq u_i^+(\boldsymbol{\xi}, \boldsymbol{\tau}) - u_i^-(\boldsymbol{\xi}, \boldsymbol{\tau})$ is displacement discontinuity across fault plane. ν is normal vector of fault plane. η is spatial coordinates. $\Sigma(\boldsymbol{\xi})$ is fault plane.

The body-force equivalent:

$$f_p^{[u]}(\eta, \boldsymbol{\tau}) = - \iint_{\Sigma} [u_i(\boldsymbol{\xi}, \boldsymbol{\tau})] \nu_j(\boldsymbol{\xi}) c_{ijpq}(\boldsymbol{\xi}) \frac{\partial \delta(\eta - \boldsymbol{\xi})}{\partial \eta_q} d\Sigma(\boldsymbol{\xi}) \quad (24)$$

(1) Prove

$$\iiint_V f^{[u]}(\eta, \boldsymbol{\tau}) dV(\eta) = 0 \quad (25)$$

Substituting (24) into (25):

$$\begin{aligned} \iiint_V f_p^{[u]}(\eta, \boldsymbol{\tau}) dV(\eta) &= \iiint_V \left(- \iint_{\Sigma} [u_i(\boldsymbol{\xi}, \boldsymbol{\tau})] \nu_j(\boldsymbol{\xi}) c_{ijpq}(\boldsymbol{\xi}) \frac{\partial \delta(\eta - \boldsymbol{\xi})}{\partial \eta_q} d\Sigma(\boldsymbol{\xi}) \right) dV(\eta) \\ &= - \iint_{\Sigma} [u_i(\boldsymbol{\xi}, \boldsymbol{\tau})] \nu_j(\boldsymbol{\xi}) c_{ijpq}(\boldsymbol{\xi}) \left(\iiint_V \frac{\partial \delta(\eta - \boldsymbol{\xi})}{\partial \eta_q} dV(\eta) \right) d\Sigma(\boldsymbol{\xi}) \end{aligned} \quad (26)$$

Evaluating the volume integral:

$$\begin{aligned} \iiint_V \frac{\partial \delta(\eta - \boldsymbol{\xi})}{\partial \eta_q} dV(\eta) &= \iiint_V \frac{\partial}{\partial \eta_q} (\delta(\eta_1 - \xi_1) \delta(\eta_2 - \xi_2) \delta(\eta_3 - \xi_3)) d\eta_q d\eta_i d\eta_j \\ &= \iint \delta(\eta_1 - \xi_1) \delta(\eta_2 - \xi_2) \delta(\eta_3 - \xi_3) \Big|_{\eta_q^-}^{\eta_q^+} d\eta_i d\eta_j \\ &= 0 \end{aligned} \quad (27)$$

The boundary term vanishes since $\delta(\eta - \boldsymbol{\xi}) = 0$ at $\eta_q \rightarrow \pm\infty$.

Substituting (27) into (26):

$$\iiint_V f^{[u]}(\eta, \boldsymbol{\tau}) dV(\eta) = 0 \quad (28)$$

(2) Prove $\iiint_V (\eta - \eta_0) \times f^{[u]}(\eta, \boldsymbol{\tau}) dV(\eta) = 0$

In component form:

$$\iiint_V (\eta - \eta_0) \times f^{[u]}(\eta, \boldsymbol{\tau}) dV(\eta) = \iiint_V \varepsilon_{mpn} (\eta_m - \eta_{0m}) f_p^{[u]}(\eta, \boldsymbol{\tau}) \mathbf{e}_n dV(\eta) \quad (29)$$

Substituting the body-force equivalent:

$$\begin{aligned} &\iiint_V \varepsilon_{mpn} (\eta_m - \eta_{0m}) f_p^{[u]}(\eta, \boldsymbol{\tau}) \mathbf{e}_n dV(\eta) \\ &= \mathbf{e}_n \iiint_V \varepsilon_{mpn} (\eta_m - \eta_{0m}) \left(- \iint_{\Sigma} [u_i(\boldsymbol{\xi}, \boldsymbol{\tau})] \nu_j(\boldsymbol{\xi}) c_{ijpq}(\boldsymbol{\xi}) \frac{\partial \delta(\eta - \boldsymbol{\xi})}{\partial \eta_q} d\Sigma(\boldsymbol{\xi}) \right) dV(\eta) \\ &= -\mathbf{e}_n \iint_{\Sigma} \varepsilon_{mpn} [u_i(\boldsymbol{\xi}, \boldsymbol{\tau})] \nu_j(\boldsymbol{\xi}) c_{ijpq}(\boldsymbol{\xi}) \left(\iiint_V (\eta_m - \eta_{0m}) \frac{\partial \delta(\eta - \boldsymbol{\xi})}{\partial \eta_q} dV(\eta) \right) d\Sigma(\boldsymbol{\xi}) \end{aligned} \quad (30)$$

Evaluating the volume integral by integration by parts:

$$\begin{aligned}
& \iiint_V (\eta_m - \eta_{0m}) \frac{\partial \delta(\eta - \boldsymbol{\xi})}{\partial \eta_q} dV(\eta) \\
&= \iiint_V (\eta_m - \eta_{0m}) \frac{\partial}{\partial \eta_q} (\delta(\eta_1 - \xi_1) \delta(\eta_2 - \xi_2) \delta(\eta_3 - \xi_3)) d\eta_q d\eta_i d\eta_j \\
&= \iint (\eta_m - \eta_{0m}) \delta(\eta_1 - \xi_1) \delta(\eta_2 - \xi_2) \delta(\eta_3 - \xi_3) \Big|_{\eta_q^-}^{\eta_q^+} d\eta_i d\eta_j \\
&\quad - \iiint_V \delta(\eta - \boldsymbol{\xi}) \frac{\partial (\eta_m - \eta_{0m})}{\partial \eta_q} d\eta_q d\eta_i d\eta_j \\
&= 0 - \iiint_V \delta(\eta - \boldsymbol{\xi}) \delta_{mq} dV(\eta) \\
&= -\delta_{mq}
\end{aligned} \tag{31}$$

Substituting (31) into (30):

$$-e_n \iint_{\Sigma} \varepsilon_{mpn} [u_i(\boldsymbol{\xi}, \boldsymbol{\tau})] \nu_j(\boldsymbol{\xi}) c_{ijpq}(\boldsymbol{\xi}) (-\delta_{mq}) d\Sigma(\boldsymbol{\xi}) = e_n \iint_{\Sigma} \varepsilon_{qpn} [u_i(\boldsymbol{\xi}, \boldsymbol{\tau})] \nu_j(\boldsymbol{\xi}) c_{ijpq}(\boldsymbol{\xi}) d\Sigma(\boldsymbol{\xi}) \tag{32}$$

Using the symmetry of elastic tensor $c_{ijpq} = c_{ijqp}$:

$$\begin{aligned}
\varepsilon_{qpn} c_{ijpq} &= \varepsilon_{qpn} \frac{c_{ijpq} + c_{ijqp}}{2} = \frac{1}{2} (\varepsilon_{qpn} c_{ijpq} + \varepsilon_{qpn} c_{ijqp}) \\
&= \frac{1}{2} (\varepsilon_{qpn} c_{ijpq} + \varepsilon_{pqn} c_{ijpq}) = \frac{1}{2} (\varepsilon_{qpn} - \varepsilon_{qpn}) c_{ijpq} = 0
\end{aligned} \tag{33}$$

Substituting (33) into (32):

$$\iiint_V (\eta - \eta_0) \times f^{[u]}(\eta, \boldsymbol{\tau}) dV(\eta) = 0 \quad \text{Done (2)} \tag{34}$$

Problem 3 Green's Function for Poisson Equation

Prove $G(\mathbf{x}; \boldsymbol{\xi}) = -\frac{1}{4\pi|\mathbf{x} - \boldsymbol{\xi}|}$ is Green's function of:

$$\nabla^2 G(\mathbf{x}; \boldsymbol{\xi}) = \delta(\mathbf{x} - \boldsymbol{\xi}) \tag{35}$$

Proof:

By Problem 4, the Green's function (41) is:

$$g(\mathbf{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = \frac{1}{4\pi c^2 |\mathbf{x} - \boldsymbol{\xi}|} \delta \left(t - \boldsymbol{\tau} - \frac{|\mathbf{x} - \boldsymbol{\xi}|}{c} \right)$$

which satisfies (42):

$$\ddot{g}(\mathbf{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = c^2 \nabla^2 g(\mathbf{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) + \delta(t - \boldsymbol{\tau}) \delta(\mathbf{x} - \boldsymbol{\xi})$$

Set the body force as:

$$f(\mathbf{x}, t) = c^2 \delta(\mathbf{x} - \boldsymbol{\xi}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} c^2 \delta(\mathbf{x}' - \boldsymbol{\xi}) \delta(\mathbf{x} - \mathbf{x}') \delta(t - t') dt' d\mathbf{x}' \tag{36}$$

Then the wave equation with this source is:

$$\ddot{u}(\mathbf{x}, t; \boldsymbol{\xi}) = c^2 \nabla^2 u(\mathbf{x}, t; \boldsymbol{\xi}) + c^2 \delta(\mathbf{x} - \boldsymbol{\xi}) \quad (37)$$

The solution by convolution with (41):

$$\begin{aligned} u(\mathbf{x}, t; \boldsymbol{\xi}) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} c^2 \delta(\mathbf{x}' - \boldsymbol{\xi}) \frac{1}{4\pi c^2 |\mathbf{x} - \mathbf{x}'|} \delta\left(t - t' - \frac{|\mathbf{x} - \mathbf{x}'|}{c}\right) dt' d\mathbf{x}' \\ &= \frac{1}{4\pi |\mathbf{x} - \boldsymbol{\xi}|} \end{aligned} \quad (38)$$

where the result is independent of c and t .

From (37), letting $c \rightarrow \infty$:

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \nabla^2\right) u = c^2 \delta(\mathbf{x} - \boldsymbol{\xi}) \quad \Leftrightarrow \quad \nabla^2(-u) = \delta(\mathbf{x} - \boldsymbol{\xi}) \quad (39)$$

Thus $-u = -\frac{1}{4\pi |\mathbf{x} - \boldsymbol{\xi}|}$ is the solution of Poisson Equation (35).

Therefore:

$$G(\mathbf{x}; \boldsymbol{\xi}) = -\frac{1}{4\pi |\mathbf{x} - \boldsymbol{\xi}|} \quad (40)$$

Problem 4. Green's Function for Wave Equation Prove:

$$g(\mathbf{x}, t; \boldsymbol{\xi}, \tau) = \frac{1}{4\pi c^2 |\mathbf{x} - \boldsymbol{\xi}|} \delta\left(t - \tau - \frac{|\mathbf{x} - \boldsymbol{\xi}|}{c}\right) \quad (41)$$

is the solution of:

$$\ddot{g}(\mathbf{x}, t; \boldsymbol{\xi}, \tau) = c^2 \nabla^2 g(\mathbf{x}, t; \boldsymbol{\xi}, \tau) + \delta(t - \tau) \delta(\mathbf{x} - \boldsymbol{\xi}) \quad (42)$$

Proof:

Let $G(p; \boldsymbol{\xi}) = \mathcal{L}(g) = \int_0^\infty g(\mathbf{x}, t; \boldsymbol{\xi}, \tau) e^{-pt} dt$.

Then:

$$\mathcal{L}(\ddot{g}) = p^2 G(p) - pg_{t=0} - \dot{g}_{t=0} = p^2 G(p) \quad (43)$$

Taking Laplace transform of (42):

$$p^2 G = c^2 \nabla^2 G + e^{-p\tau} \delta(\mathbf{x} - \boldsymbol{\xi}) \quad (44)$$

Let $r = |\mathbf{x} - \boldsymbol{\xi}|$. By spherical symmetry, G is independent of θ and φ :

$$\nabla^2 G = \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta}\right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}\right) G = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial G}{\partial r}\right) = \frac{1}{r} \frac{\partial^2(rG)}{\partial r^2} \quad (45)$$

For $r > 0$, (44) becomes:

$$p^2 G = \frac{c^2}{r} \frac{\partial^2(rG)}{\partial r^2} \Rightarrow \frac{p^2}{c^2}(rG) = \frac{\partial^2(rG)}{\partial r^2} \quad (46)$$

The general solution of (46):

$$G = \frac{A}{r} e^{\pm \frac{p}{c} r} \quad (47)$$

Since $\text{Re}(p) > 0$, $c > 0$, $r > 0$, for G to be finite as $r \rightarrow \infty$:

$$G = \frac{A}{r} e^{-\frac{p}{c}r} \quad (48)$$

To determine A , take the open ball V_ε centered at $r = 0$ with radius ε :

$$\iiint_{V_\varepsilon} p^2 G \, dV = c^2 \iiint_{V_\varepsilon} \nabla^2 G \, dV + \iiint_{V_\varepsilon} e^{-p\tau} \delta(r) \, dV \quad (49)$$

For the first term on the L.H.S. of (49):

$$\begin{aligned} \iiint_{V_\varepsilon} p^2 G \, dV &= p^2 \int_0^{2\pi} \int_0^\pi \int_0^\varepsilon \frac{A}{r} e^{-\frac{p}{c}r} r^2 \sin \theta \, dr \, d\theta \, d\varphi \\ &= 4\pi p^2 A \int_0^\varepsilon r e^{-\frac{p}{c}r} \, dr = 4\pi p^2 A (\varepsilon - 0) \xi e^{-\frac{p}{c}\xi} \quad (\xi \in [0, \varepsilon] \text{ by mean value theorem}) \\ &= 0 \quad (\text{as } \varepsilon \rightarrow 0) \end{aligned} \quad (50)$$

For the first term on the R.H.S. of (49), by divergence theorem:

$$c^2 \iiint_{V_\varepsilon} \nabla^2 G \, dV = c^2 \oint_{S_\varepsilon} \nabla G \cdot \hat{n} \, dS \quad (51)$$

In spherical coordinates:

$$\nabla = \hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{e}_\varphi \frac{1}{r \sin \theta} \frac{\partial}{\partial \varphi} \quad (52)$$

From (48):

$$\nabla G = \hat{e}_r \frac{\partial}{\partial r} \left(\frac{A}{r} e^{-\frac{p}{c}r} \right) = \hat{e}_r A \frac{-\frac{p}{c} r e^{-\frac{p}{c}r} - e^{-\frac{p}{c}r}}{r^2} \quad (53)$$

Evaluating (51):

$$\begin{aligned} c^2 \oint_{S_\varepsilon} \nabla G \cdot \hat{n} \, dS &= \int_0^{2\pi} \int_0^\pi c^2 \frac{A}{r^2} \left(-\frac{p}{c} r e^{-\frac{p}{c}r} - e^{-\frac{p}{c}r} \right) r^2 \sin \theta \, d\theta \, d\varphi \\ &= 4\pi c^2 A \left(-\frac{p}{c} \varepsilon e^{-\frac{p}{c}\varepsilon} - e^{-\frac{p}{c}\varepsilon} \right) \\ &= -4\pi c^2 A \quad (\text{as } \varepsilon \rightarrow 0) \end{aligned} \quad (54)$$

For the source and second term on the R.H.S. of (49):

$$\iiint_{V_\varepsilon} e^{-p\tau} \delta(r) \, dV = e^{-p\tau} \quad (55)$$

(54), Substituting (50), and (55) into (49):

$$0 = -4\pi c^2 A + e^{-p\tau} \Rightarrow A = \frac{e^{-p\tau}}{4\pi c^2} \quad (56)$$

Therefore from (48) and (56):

$$G = \frac{1}{4\pi c^2 r} e^{-p(\frac{r}{c} + \tau)} \quad (57)$$

By the inverse Laplace transform (Bromwich integral):

$$g = \frac{1}{2\pi i} \lim_{\delta \rightarrow \infty} \int_{s-i\delta}^{s+i\delta} G e^{pt} dp \quad (58)$$

Since there is no pole in the right half-plane (see Figure), we can evaluate (58):

$$\begin{aligned} g &= \frac{1}{2\pi i} \lim_{\delta \rightarrow \infty} \int_{s-i\delta}^{s+i\delta} \frac{e^{-p(\frac{r}{c} + \tau - t)}}{4\pi c^2 r} dp \\ &= \frac{e^{-s(\frac{r}{c} + \tau - t)}}{4\pi c^2 r} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\delta(\frac{r}{c} + \tau - t)} d\delta \\ &= \frac{e^{-s(\frac{r}{c} + \tau - t)}}{4\pi c^2 r} \delta\left(t - \frac{r}{c} - \tau\right) \end{aligned} \quad (59)$$

(by the definition of Dirac delta function, $f(x)\delta(x-a) = f(a)\delta(x-a)$)

$$= \frac{\delta\left(t - \frac{r}{c} - \tau\right)}{4\pi c^2 r} \quad (60)$$

Therefore from (60):

$$g(\mathbf{x}, t; \boldsymbol{\xi}, \tau) = \frac{\delta\left(t - \tau - \frac{|\mathbf{x} - \boldsymbol{\xi}|}{c}\right)}{4\pi c^2 |\mathbf{x} - \boldsymbol{\xi}|} \quad \text{Done (4)} \quad (61)$$

is the solution of (42).

Problem 5 Helmholtz Potentials

According to Box 4.2, a given vector function $\mathbf{f}(\mathbf{x})$ can be decomposed into:

$$\mathbf{f}(\mathbf{x}) = \nabla \Phi(\mathbf{x}) + \nabla \times \Psi(\mathbf{x}) \quad (62)$$

Potentials $\Psi(\mathbf{x})$ and $\Phi(\mathbf{x})$ can be obtained by generating function $\mathbf{W}(\mathbf{x})$:

$$\Psi(\mathbf{x}) = -\nabla \times \mathbf{W}(\mathbf{x}) \quad \text{and} \quad \Phi(\mathbf{x}) = \nabla \cdot \mathbf{W}(\mathbf{x}) \quad (63)$$

where:

$$\mathbf{W}(\mathbf{x}) = - \iiint_V \frac{\mathbf{f}(\boldsymbol{\xi})}{4\pi |\mathbf{x} - \boldsymbol{\xi}|} dV(\boldsymbol{\xi}) \quad (64)$$

$$(1) \mathbf{f}(\mathbf{x}) = X_i(t) \mathbf{e}_i \delta(\mathbf{x})$$

The generating function:

$$\mathbf{W}(\mathbf{x}) = - \iiint_V \frac{X_i(t) \mathbf{e}_i \delta(\boldsymbol{\xi})}{4\pi |\mathbf{x} - \boldsymbol{\xi}|} dV(\boldsymbol{\xi}) = - \frac{X_i(t) \mathbf{e}_i}{4\pi |\mathbf{x}|} \quad (65)$$

Using $\frac{\partial |\mathbf{x}|}{\partial x_m} = \frac{x_m}{|\mathbf{x}|}$ and $\frac{\partial}{\partial x_m} \left(\frac{1}{|\mathbf{x}|} \right) = -\frac{x_m}{|\mathbf{x}|^3}$:

The scalar potential:

$$\Phi(\mathbf{x}) = \nabla \cdot \mathbf{W}(\mathbf{x}) = -\frac{\partial}{\partial x_m} \left(\frac{X_m(t)}{4\pi |\mathbf{x}|} \right) = -\frac{X_m(t)}{4\pi} \left(-\frac{x_m}{|\mathbf{x}|^3} \right) = \frac{X_m(t) x_m}{4\pi |\mathbf{x}|^3} \quad (66)$$

The vector potential:

$$\begin{aligned}\Psi(\mathbf{x}) &= -\nabla \times \mathbf{W}(\mathbf{x}) = \nabla \times \left(\frac{X_i(t)\mathbf{e}_i}{4\pi|\mathbf{x}|} \right) = \varepsilon_{mnp} \frac{\partial}{\partial x_m} \left(\frac{X_n(t)}{4\pi|\mathbf{x}|} \right) \mathbf{e}_p \\ &= -\varepsilon_{mnp} \frac{x_m}{|\mathbf{x}|^3} \frac{X_n(t)}{4\pi} \mathbf{e}_p = -\frac{1}{4\pi} \frac{(\mathbf{x} \times \mathbf{X}(t))}{|\mathbf{x}|^3}\end{aligned}\quad (67)$$

$$(2) \mathbf{f}(\mathbf{x}) = -M_{ij}(t)\mathbf{e}_i \frac{\partial}{\partial x_j} \delta(\mathbf{x})$$

The generating function:

$$\mathbf{W}(\mathbf{x}) = - \iiint_V \frac{-M_{ij}(t)\mathbf{e}_i \frac{\partial}{\partial \xi_j} \delta(\boldsymbol{\xi})}{4\pi|\mathbf{x} - \boldsymbol{\xi}|} dV(\boldsymbol{\xi}) \quad (68)$$

Using integration by parts and $\frac{\partial}{\partial \xi_j} \left(\frac{1}{|\mathbf{x} - \boldsymbol{\xi}|} \right) = \frac{x_j - \xi_j}{|\mathbf{x} - \boldsymbol{\xi}|^3}$:

$$\begin{aligned}\mathbf{W}(\mathbf{x}) &= \mathbf{e}_i \frac{M_{ij}(t)}{4\pi} \iiint_V \frac{\frac{\partial}{\partial \xi_j} \delta(\boldsymbol{\xi})}{|\mathbf{x} - \boldsymbol{\xi}|} dV(\boldsymbol{\xi}) \\ &= \mathbf{e}_i \frac{M_{ij}(t)}{4\pi} \left(\iint_S \frac{\delta(\boldsymbol{\xi})}{|\mathbf{x} - \boldsymbol{\xi}|} \Big|_{\xi_j^-}^{\xi_j^+} d\xi_p d\xi_q - \iiint_V \delta(\boldsymbol{\xi}) \frac{\partial}{\partial \xi_j} \left(\frac{1}{|\mathbf{x} - \boldsymbol{\xi}|} \right) dV \right) \\ &= \mathbf{e}_i \frac{M_{ij}(t)}{4\pi} \left(- \iiint_V \delta(\boldsymbol{\xi}) \frac{x_j - \xi_j}{|\mathbf{x} - \boldsymbol{\xi}|^3} dV \right) \\ &= -\mathbf{e}_i \frac{M_{ij}(t)}{4\pi} \frac{x_j}{|\mathbf{x}|^3}\end{aligned}\quad (69)$$

The scalar potential:

$$\begin{aligned}\Phi(\mathbf{x}) &= \nabla \cdot \mathbf{W}(\mathbf{x}) = \frac{\partial}{\partial x_n} \left(-\frac{M_{nj}(t)}{4\pi} \frac{x_j}{|\mathbf{x}|^3} \right) \\ &= -\frac{M_{nj}(t)}{4\pi} \left(\frac{\delta_{jn}|\mathbf{x}|^3 - 3x_j|\mathbf{x}|^2 \frac{x_n}{|\mathbf{x}|}}{|\mathbf{x}|^6} \right) \\ &= -\frac{M_{nj}(t)}{4\pi} \left(\frac{\delta_{jn} - 3x_j x_n / |\mathbf{x}|^2}{|\mathbf{x}|^3} \right) \\ &= -\frac{1}{4\pi} \left(\frac{M_{nn}}{|\mathbf{x}|^3} - \frac{3M_{nj}x_n x_j}{|\mathbf{x}|^5} \right)\end{aligned}\quad (70)$$

The vector potential:

$$\begin{aligned}\Psi(\mathbf{x}) &= -\nabla \times \mathbf{W}(\mathbf{x}) = -\varepsilon_{mnp} \frac{\partial}{\partial x_m} \left(-\frac{M_{nj}(t)}{4\pi} \frac{x_j}{|\mathbf{x}|^3} \right) \mathbf{e}_p \\ &= \frac{M_{nj}(t)}{4\pi} \varepsilon_{mnp} \frac{\partial}{\partial x_m} \left(\frac{x_j}{|\mathbf{x}|^3} \right) \mathbf{e}_p \\ &= \frac{M_{nj}(t)}{4\pi} \varepsilon_{mnp} \left(\frac{\delta_{jm}|\mathbf{x}|^3 - 3x_j|\mathbf{x}|^2 \frac{x_m}{|\mathbf{x}|}}{|\mathbf{x}|^6} \right) \mathbf{e}_p \\ &= \frac{M_{nj}(t)}{4\pi} \varepsilon_{mnp} \left(\frac{\delta_{jm} - 3x_j x_m / |\mathbf{x}|^2}{|\mathbf{x}|^3} \right) \mathbf{e}_p \\ &= \left(\frac{\varepsilon_{mnp} M_{nm}(t)}{4\pi|\mathbf{x}|^3} - \frac{3}{4\pi|\mathbf{x}|^5} \varepsilon_{mnp} M_{nj}(t) x_j x_m \right) \mathbf{e}_p\end{aligned}\quad (71)$$

$$(3) \mathbf{f}(\mathbf{x}) = -M_0(t)\nabla\delta(\mathbf{x})$$

The generating function:

$$\begin{aligned} \mathbf{W}(\mathbf{x}) &= -\iiint_V \frac{-M_0(t)\nabla_{\xi}\delta(\xi)}{4\pi|\mathbf{x}-\xi|}dV = e_p \frac{M_0(t)}{4\pi} \iiint_V \frac{1}{|\mathbf{x}-\xi|} \frac{\partial\delta(\xi)}{\partial\xi_p} d\xi_p d\xi_m d\xi_n \\ &= e_p \frac{M_0(t)}{4\pi} \left(\iint \frac{\delta(\xi)}{|\mathbf{x}-\xi|} \Big|_{\xi_p^-}^{\xi_p^+} d\xi_m d\xi_n - \iiint_V \delta(\xi) \frac{\partial}{\partial\xi_p} \left(\frac{1}{|\mathbf{x}-\xi|} \right) dV \right) \\ &= -e_p \frac{M_0(t)}{4\pi} \iiint_V \delta(\xi) \frac{x_p - \xi_p}{|\mathbf{x}-\xi|^3} dV(\xi) \\ &= -e_p \frac{M_0(t)x_p}{4\pi|\mathbf{x}|^3} \\ &= -\frac{M_0(t)}{4\pi} \frac{\mathbf{x}}{|\mathbf{x}|^3} \end{aligned} \quad (72)$$

The scalar potential:

$$\Phi(\mathbf{x}) = \nabla \cdot \mathbf{W}(\mathbf{x}) = \nabla \cdot \left(-\frac{M_0(t)}{4\pi} \frac{\mathbf{x}}{|\mathbf{x}|^3} \right) = \frac{M_0(t)}{4\pi} \nabla \cdot \left(-\frac{\mathbf{x}}{|\mathbf{x}|^3} \right) = \frac{M_0(t)}{4\pi} \nabla^2 \left(\frac{1}{|\mathbf{x}|} \right) \quad (73)$$

By Problem 3, $\nabla^2 \left(-\frac{1}{4\pi|\mathbf{x}-\xi|} \right) = \delta(\mathbf{x}-\xi)$. With $\xi = 0$:

$$\nabla^2 \left(\frac{1}{|\mathbf{x}|} \right) = -4\pi\delta(\mathbf{x}) \quad (74)$$

Substituting (74) into (73):

$$\Phi(\mathbf{x}) = \frac{M_0(t)}{4\pi} (-4\pi\delta(\mathbf{x})) = -M_0(t)\delta(\mathbf{x}) \quad (75)$$

The vector potential:

$$\begin{aligned} \Psi(\mathbf{x}) &= -\nabla \times \mathbf{W}(\mathbf{x}) = -\varepsilon_{mnp} \frac{\partial}{\partial x_m} \left(-\frac{M_0(t)x_n}{4\pi|\mathbf{x}|^3} \right) \mathbf{e}_p \\ &= \frac{M_0(t)}{4\pi} \varepsilon_{mnp} \frac{\partial}{\partial x_m} \left(\frac{x_n}{|\mathbf{x}|^3} \right) \mathbf{e}_p \\ &= \frac{M_0(t)}{4\pi} \varepsilon_{mnp} \left(\frac{\delta_{nm}}{|\mathbf{x}|^3} - \frac{3x_n x_m}{|\mathbf{x}|^5} \right) \mathbf{e}_p \\ &= e_p \frac{M_0(t)}{4\pi} \left(\frac{\varepsilon_{mmp}}{|\mathbf{x}|^3} - \frac{3\varepsilon_{mnp}x_n x_m}{|\mathbf{x}|^5} \right) \mathbf{e}_p \end{aligned} \quad (76)$$

Since $\varepsilon_{mmp} = 0$ (repeated index) and $\varepsilon_{mnp}x_n x_m = \varepsilon_{mnp}(x_n x_m + x_m x_n)/2 = (\varepsilon_{mnp} + \varepsilon_{nmp})x_n x_m/2 = 0$ by antisymmetry:

$$\Psi(\mathbf{x}) = 0 \quad (77)$$